

Lesson 2

Understanding Virtualization and Cloud Computing

Topic 2A

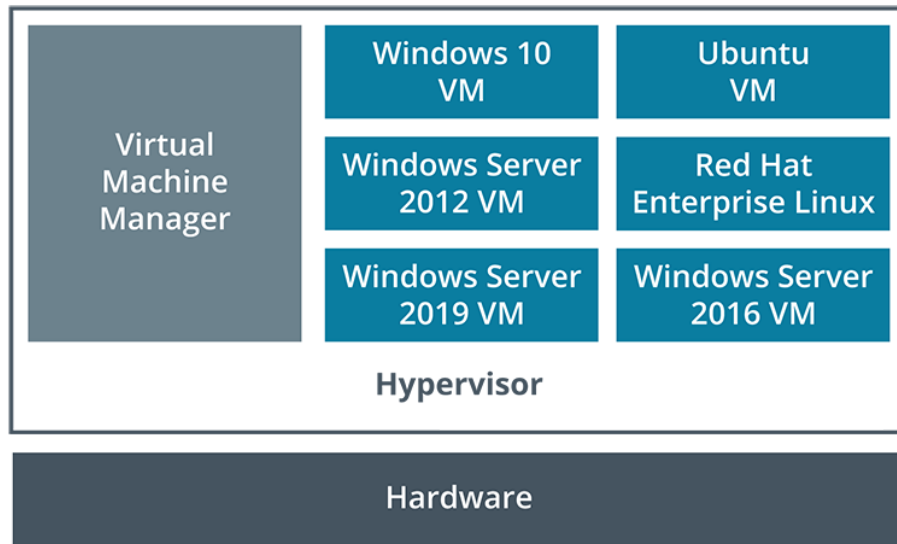
Understand Virtualization Concepts

Exam Objectives Covered

- 2.5 Summarize the purpose and operation of virtualization

Virtualization Concepts

- Virtualization
- Redundancy and High Availability
- Host system versus guest system
 - Type 1
 - Type 2



Hardware layer, a hypervisor, guest VMs, and a virtual machine manager.

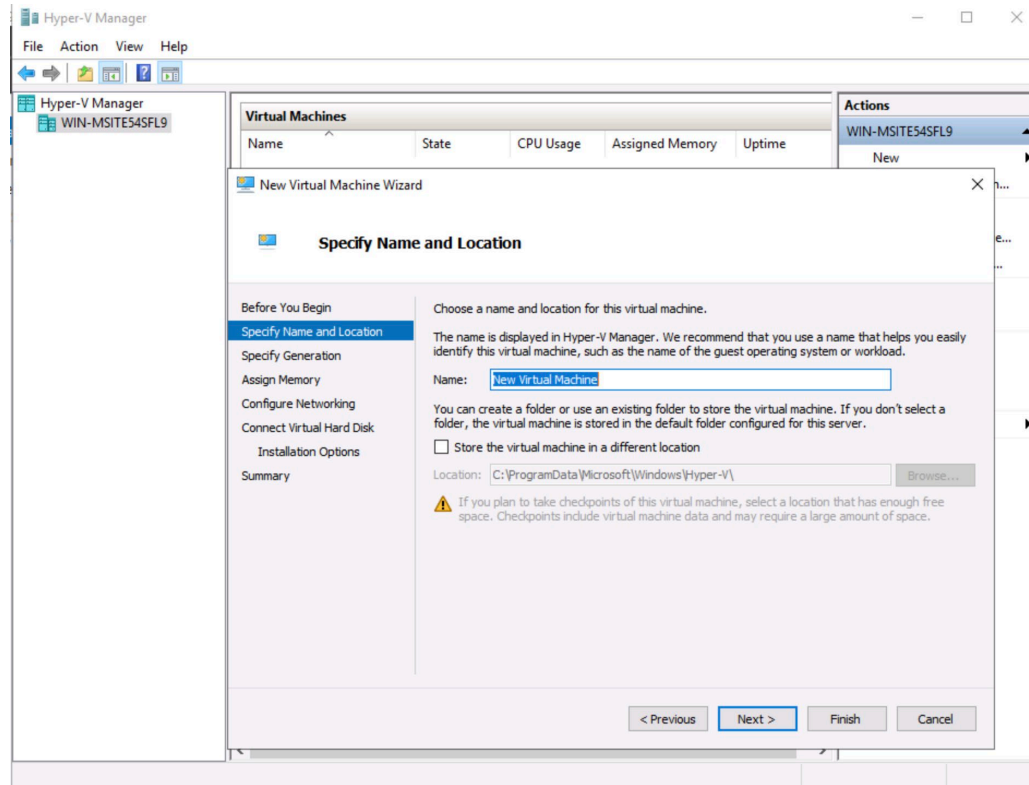
Manage Virtualization Resources

- Overprovision
- Over-allocate resources

Virtual Networking

- Options for network access
 - None
 - Local
 - Host-only
 - Bridged
- Network address translation (NAT)
- Management interfaces for VMs

Windows Server Hyper-V console



Screenshot courtesy of Microsoft.

Review Activity: Virtualization Concepts



Review Activity

Topic 2B

Understanding Cloud Concepts

Exam Objectives Covered

- 2.5 Summarize the purpose and operation of virtualization

Cloud Characteristics

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

Cloud Deployment Models

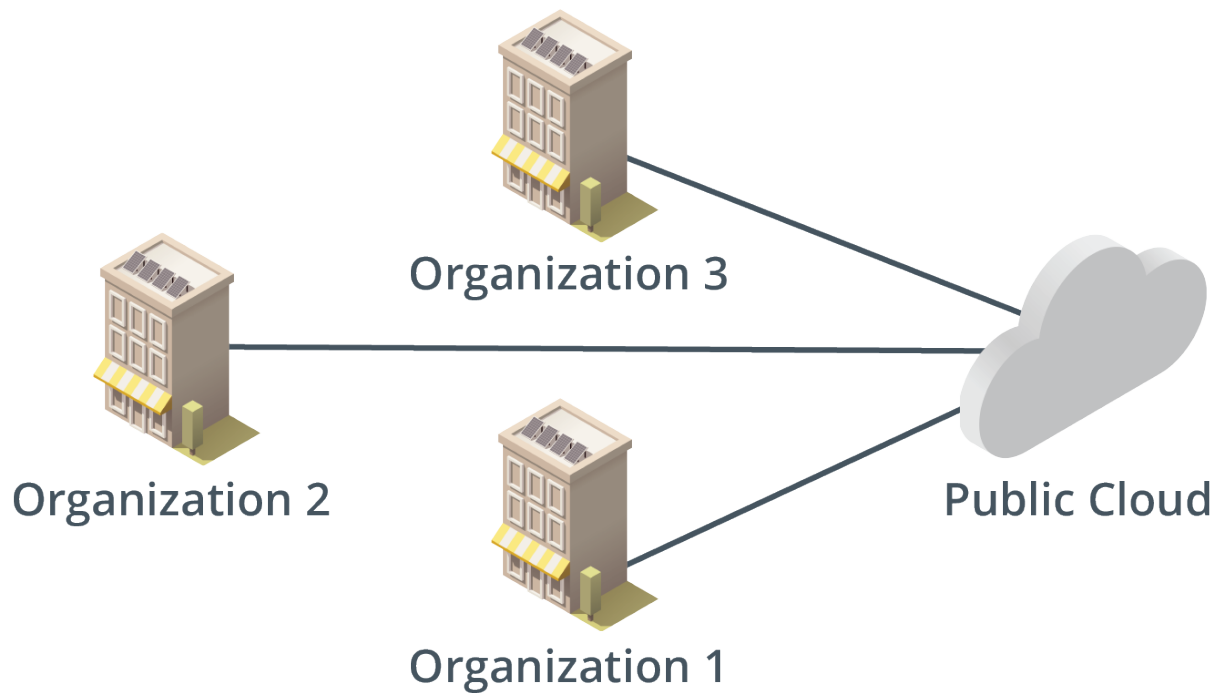
- Private cloud
- Public cloud
- Community cloud
- Hybrid cloud

Private Cloud



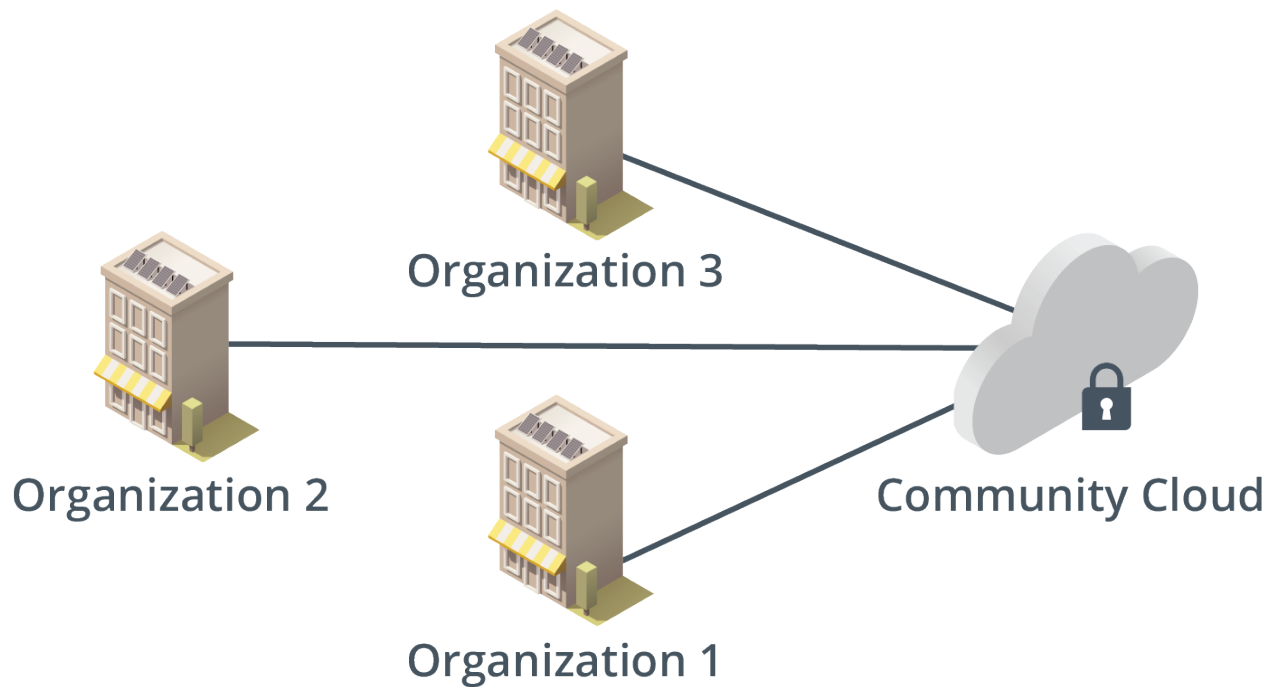
Private cloud model. (Images © 123RF.com)

Public Cloud



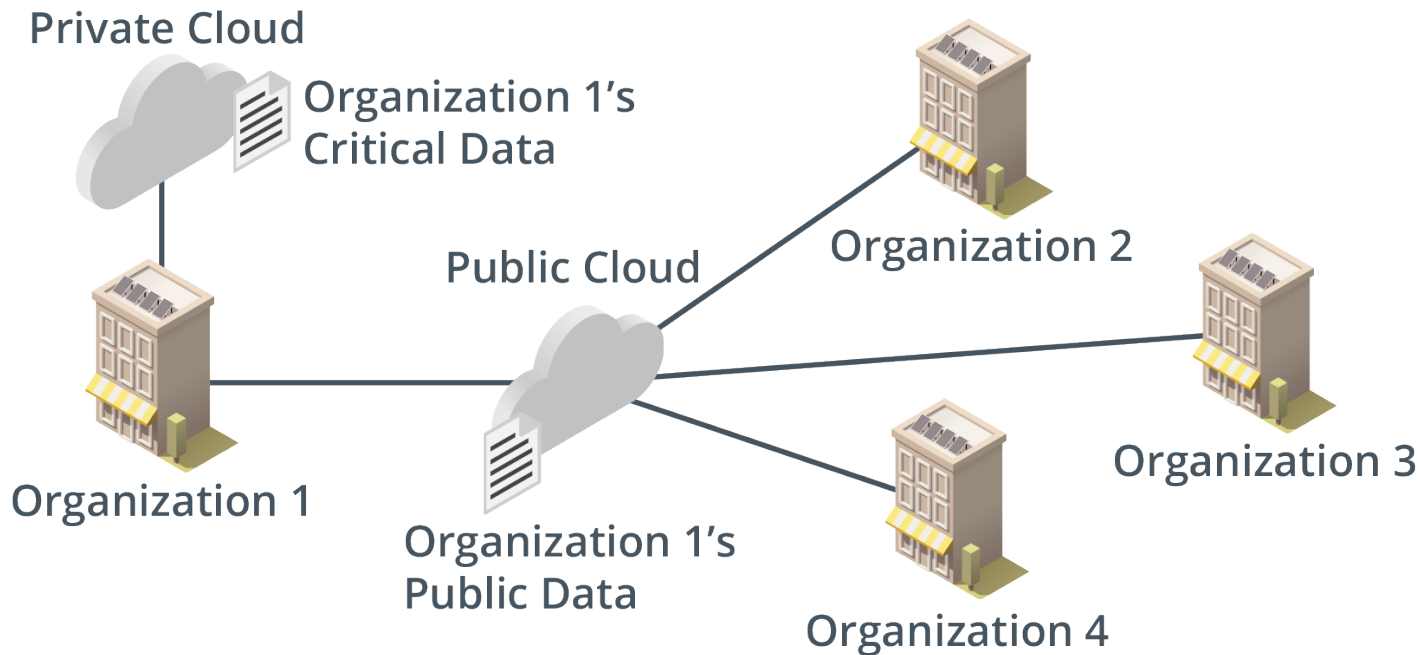
*Public cloud model, showing three organizations sharing one publicly accessible cloud.
(Images © 123RF.com)*

Community Cloud



*Community cloud model, showing three organizations sharing a secured cloud.
(Images © 123RF.com)*

Hybrid Cloud



Hybrid cloud model. (Images © 123RF.com)

Cloud Service Models

Software as a Service (SaaS)	Hosted Applications	 Office 365	 Salesforce	 GMail
	Infrastructure Software	 Oracle	 MySQL	 Java
	Operating Systems	 Windows	 Linux	
Platform as a Service (PaaS)	Virtualization	 Xen	 VMWare	 Hyper-V
	Servers and Server Hardware	 Servers	 CPU	 Memory
				 Storage
	Networking	 Firewalls	 Routers	 Switches
	Data Center Electrical and Mechanical Operations	 Power	 Cooling	 Physical Security

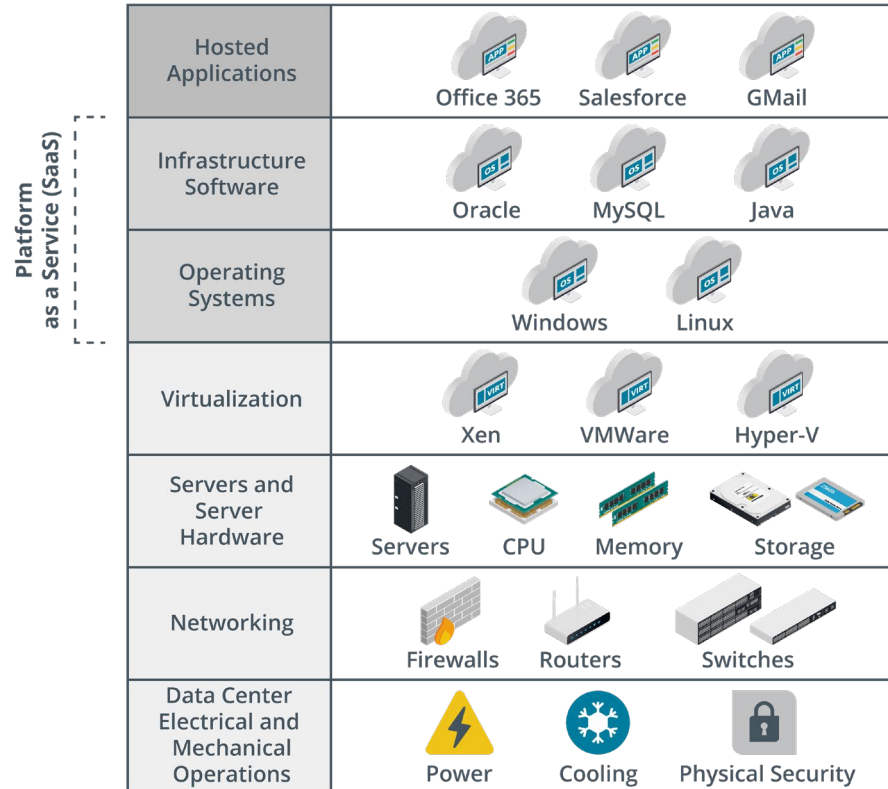
Three cloud service models. (Images © 123RF.com)

SaaS

Software as a Service (SaaS)	Hosted Applications	Office 365	Salesforce	GMail
	Infrastructure Software	Oracle	MySQL	Java
	Operating Systems	Windows	Linux	
	Virtualization	Xen	VMWare	Hyper-V
	Servers and Server Hardware	Servers	CPU	Memory
	Networking	Firewalls	Routers	Switches
	Data Center Electrical and Mechanical Operations	Power	Cooling	Physical Security

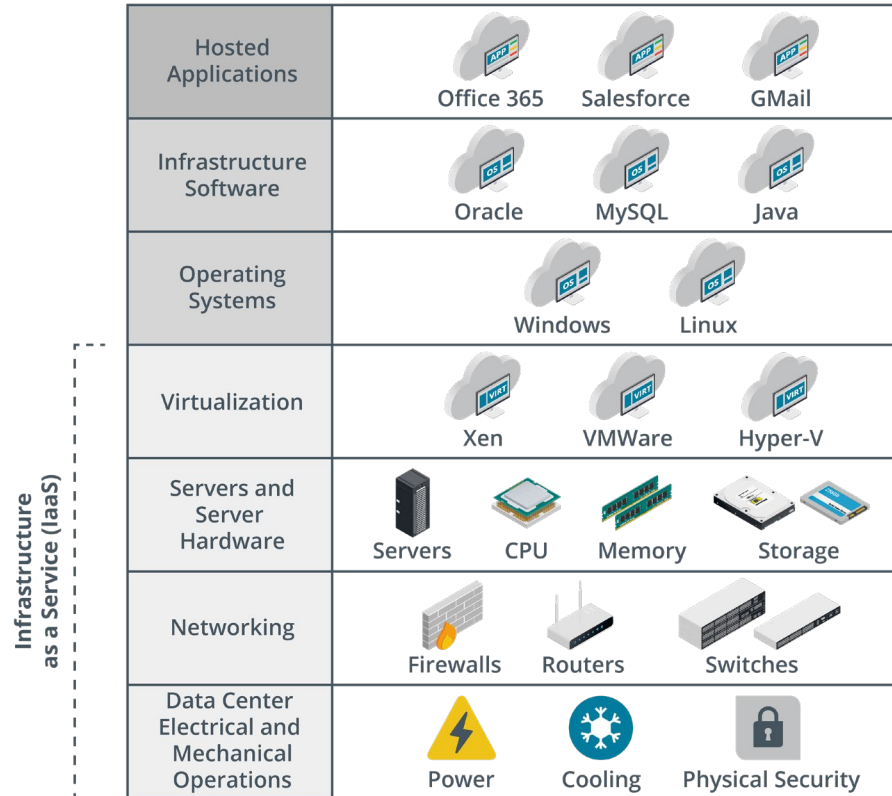
SaaS. (Images © 123RF.com)

PaaS



PaaS. (Images © 123RF.com)

IaaS



IaaS. (Images © 123RF.com)

Cloud Service Providers

- Most common
 - Amazon--AWS
 - Microsoft—Azure
 - Google—GCP
- The Shared Security Model

Guidelines for Understanding Cloud Concepts

- Five characteristics of cloud computing
- Four deployment models
- Three service models
- Three major CSPs
- Shared security model division of responsibility

Review Activity: Cloud Computing



Review Activity

Topic 2C

Understand On-Premises Versus Cloud Deployments

Exam Objectives Covered

- 2.5 Summarize the purpose and operation of virtualization

On-Premises, Physical Servers

- Organization maintains servers
 - Provisioning
 - Administration
 - Security
 - Troubleshooting
 - Purchasing

On-Premises, Virtual Servers

- Same benefits as physical servers, with greater:
 - Scalability
 - Disaster Recovery
 - High Availability

Cloud, Virtual Servers

- No direct support costs
- Increased disaster recovery and high availability
- Simplified licensing—pay-as-you-go model
- Less control

On-Premises versus Cloud Deployments

On-Premises Advantages

- Organization retains control of deployment environment, administration, management and data
- Data close to users
- Application customization
- Specialized hardware

On-Premises Disadvantages

- Organization bears cost of hardware, installation, maintenance, data security and personnel
- Higher CapEx
- Less hardware flexibility
- Difficult to scale
- Slower time to deployment

On-Premises versus Cloud Deployments (slide 2)

Cloud Advantages

- Organization pays only for consumed cloud resources and retains control of most service administration and management
- Org does not bear cost to support
- Many CSPs certified for PII, PCI DSS, and many other standards
- Simplified licensing and pay-as-you-go model
- Data can be replicated worldwide

Cloud Disadvantages

- Organization doesn't retain control of the deployment environment
- Org may lose data access due to internet connectivity issues or CSP downtime
- May increase OpEx
- Data may be maintained farther from users

On-Premises versus Cloud Deployments (slide 3)

- Small and mid-sized organizations may prefer low cost and flexible cloud
- Organizations with unique or high-security requirements may prefer on-premises deployment for full control of data and services

Review Activity: Deployment Models



Review Activity

Lesson 2

Summary

